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PROJECT REPORT

DYC CODING CAMPUS – A WEB BASED ONLINE COMPILER PLATFORM

COMPANY : TECH S CUBE

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ABSTRACT

The DYC CODING CAMPUS – A WEB-BASED ONLINE COMPILER PLATFORM is a modern web application that enables users to write, compile, execute, and manage programs directly through a web browser without installing language-specific software or development tools on their local systems. The platform supports multiple programming languages such as Python, JavaScript, C, C++, and Java through secure containerized execution environments.

The system utilizes Docker containerization technology to isolate code execution processes and eliminate dependency conflicts, thereby ensuring security, scalability, and reliability. Users can create accounts, manage source code files, execute programs, and view execution history through an intuitive web interface.

The platform is developed using React.js for the frontend, FastAPI for backend services, MongoDB for data storage, and Docker for isolated code execution. Additional modules include authentication, file management, execution monitoring, activity logging, and administrative controls.

The proposed solution provides an accessible, secure, and efficient environment for students, programmers, and educators to practice, test, and evaluate programming concepts online. The architecture is designed to support future enhancements such as AI- Assisted code suggestions, coding assessments, online examinations, code sharing, and collaborative programming environments.